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Date

Biometrical Stats & Hypothesis Testing

Hypothesis Testing and its mechanism:

⇒ Hyp. Testing Mechanism:

- ① write Null Hypothesis (H_0) → the assumption you are beginning with
- ② write Alternate Hypothesis (H_1) → opp. of Null hypothesis
- ③ Perform statistical tests → collect proof

P-value & Hypothesis testing:

⇒ P-value: It's a number, calculated from statistical test, that describes how likely you are to have found a particular set of observations if the null hypo. were true.

⇒ Hypothesis testing: Coin is fair or not

a) Null Hypo:

H_0 : Coin is fair

b) Alt. Hypo:

H_1 : coin is not fair

c) Exp: tossed for 100 times

(इस पेड)

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सुपर स्टील जलवायु, सुपर मजबूत धातु

0 1 0 7 0 3 0 4 0 5 0 6 0 7 0 8 0

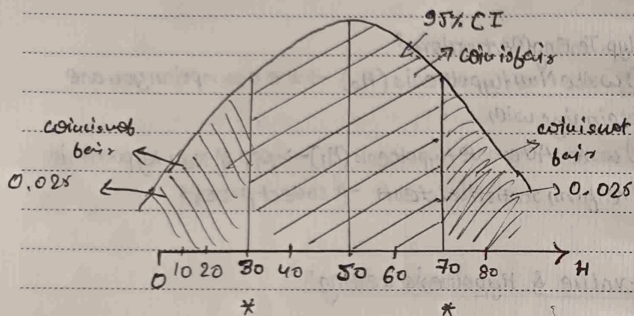


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d) Significance value :

$$\alpha = 0.05$$

(confidence inter) C.I. = $1 - 0.05 = 0.95$



If $p < \text{Significance value} \rightarrow \text{Reject } H_0$

Else

Fail to reject H_0

e) Conclusion :

Student T distribution :

$\Rightarrow$ We use it when we don't have popⁿ std. deviation.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

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$\Rightarrow$ Degree of freedom :

$$dof = n - 1$$

Z-test and Hypothesis testing :

$\Rightarrow$ Statistical analysis :

- a) Z-test } average
- b) t-test } average
- c) Chi-square $\Rightarrow$ categorical data
- d) Anova $\Rightarrow$ variables

$\Rightarrow$ Z-test :

① Popⁿ std. deviation is needed

② $n > 30$

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$Z_{score} = \frac{\bar{x} - \mu}{\sigma}$$

T-test with t Test Hypothesis testing :

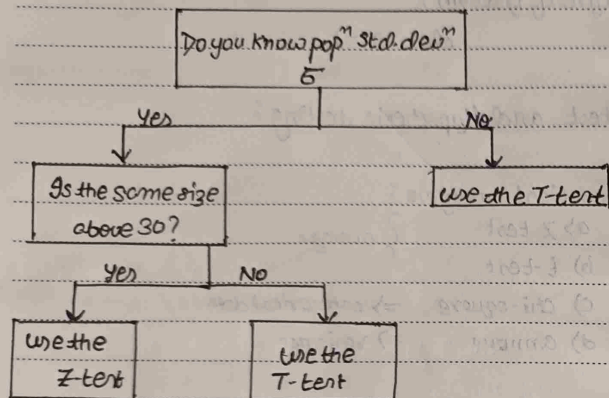
When to use T-test & Z-test :

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Type 1 & Type 2 errors:

⇒ Type 1 error: we reject the Null hypo. when in reality it's true.

⇒ Type 2 error: we accept the " " " " it's false

⇒ Good or correct: we accept the " " " " it's true

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Bayes Theorem:

⇒ Bayes Statistics: It's an approach to data analysis & parameter estimation based on Bayes Theorem.

Ind. events

- Ⓐ Rolling a dice
- Ⓑ Tossing a coin

Dependent events

- Ⓐ Taking card if 1 comes and die
- Ⓑ Wearing red shirt if it rains

$$\Rightarrow P(R \text{ and } Y) = P(R) * P(Y/R)$$

$$\Rightarrow P(A \text{ and } B) = P(B \text{ and } A)$$

$$\Rightarrow P(A) * P(B/A) = P(B) * P(A/B)$$

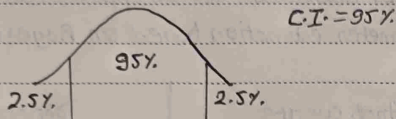
$$\Rightarrow \boxed{P(B/A) = \frac{P(B) * P(A/B)}{P(A)}} \rightarrow \text{Bayes' theorem}$$

where, $P(A/B)$ = Prob of A given B is true
 $P(A), P(B) \rightarrow$ ind. prob.

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Confidence Interval and Margin of Error:

⇒



⇒ Confidence Interval:

Point Estimate $\pm$ Margin of Error

Z-test: $\bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$

$(9000/9) - (8000/9)$

$(8/9) \times (9) = (8/9) \times (9) = 8$

$(9000/9) - (8000/9) = 8$

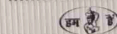
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बालमुकुन्द सुपर स्कूल
पुरा शिक्षा अकादमी, पुरा मन्दाप बजार

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